502.04.05 Final Quality Management Testing Reports - Prior to the issuance of a Completion Certificate, the Contractor shall provide the Ministry Representative with:

- a summary of all aggregate quality test results;
- copies of all quality test results for asphalt mix properties and compaction; and
- copies of all quality charts.

502.04.06 Minimum Acceptable Construction Practices - Good construction practices shall be considered as standard construction procedures to be followed and shall include but not be limited to the procedures described below.

Tack Coating - The surface to be tack coated shall be cleaned of dirt or other foreign material.

Vertical Surfaces – Vertical surfaces of roadway appurtenances shall be hand sprayed to the top of the new pavement only.

Tack Coat – Application – Tack coat shall not be applied on sections of roadway longer than will be covered by one day of asphalt plant production.

Stockpiles – Stockpiles of different types of material shall be located and constructed in such a manner as to prevent intermingling of the types and to prevent segregation.

Haul Vehicles – Lubrication of the truck boxes with diesel fuel will not be permitted. All vehicles shall have adequately insulated truck boxes and shall be equipped with an insulating tarpaulin of such size as to completely protect the asphalt mix.

Longitudinal Joints – Longitudinal joints in the top lift of asphalt pavement will only be permitted where lane dividing lines are to be painted. Longitudinal joints in a lift shall be offset at least 150 mm from joints in the underlying lift.

Longitudinal Edge – Where an unprotected longitudinal edge has been damaged by traffic or equipment it shall be trimmed to provide a vertical abutting face.

Contact Edge – The contact edge of abutting cooled asphalt pavement shall be thoroughly painted with a uniform coat of emulsified asphalt.

Paving Adjoining Mats – When paving an adjoining parallel mat there shall be at least a 50-mm overlap onto the previously paved surface. This overlap shall be properly constructed to form a homogeneous bond between the two mats.

Raking – Any material that is raked from the joint shall not be placed on the new mat or placed in or in front of the paver. The Contractor shall dispose of the material in an environmentally acceptable manner.

Discontinued Paving – When paving is discontinued in any lane on a traffic bearing roadway, the asphalt pavement shall be tapered down at a minimum slope of 25:1. When paving resumes this letdown shall be cut back so as to form a vertical face that matches the required depth being paved. Material removed from the taper shall be disposed of in an environmentally friendly manner.

Rollers – Rollers shall normally operate with the drive wheel nearest the paver and at a speed not in excess of 8 km/h. They shall not be driven on to or off, the unprotected longitudinal edge of the unfinished mat and shall not be allowed to park on the mat prior to complete cooling.

502.05 Supply Of Aggregates, Asphalt Materials, Asphalt Mix Design And Job Mix Formula

502.05.01 Work in Ministry Pits or Quarries - When operating in a Ministry pit or quarry, the Contractor shall comply with all provisions of DBSS 145.26.

502.05.02 Supply of Aggregates, Aggregate Production and Characteristics -The Contractor shall not produce paving aggregate until the Contractor has received written notification that their Quality Management Plan is in accordance with DBSS 502.04.02, and has in place testing facilities for aggregate production that are in accordance with the Quality Management Plan.

For the production of Asphalt Mix Aggregate, within Ministry pits, the Contractor shall provide crushing equipment such that all aggregate which will pass through 375 mm x 450 mm slotted openings shall be used for the production of crushed aggregate. Rocks, which will not pass through these openings, shall be stockpiled to the satisfaction of the Ministry Representative. Crushing and screening equipment shall be provided with adequate facilities and capacity to be able to bleed off reject aggregate and remove any excess fine aggregate, dust or objectionable aggregate coatings, to make it generally acceptable for use. No portion of the products from crushing or screening plants that can be used shall be wasted, but shall be stockpiled or used to the satisfaction of the Ministry Representative.

Where the Ministry has available any test result information on the properties shown in Table 502-B, for a Ministry pit or other source, the Ministry Representative will upon request, provide that information to the Contractor. Otherwise, sampling and testing to determine and demonstrate the compliance of paving aggregate with the requirements of this Section shall be the responsibility of the Contractor.

Paving aggregates shall meet the following requirements:

a) Coarse Aggregates

i) shall be all mineral matter retained on the sieve designated in the test procedures for each individual test.

ii) shall consist of crushed stone, crushed gravel, or combination thereof, or materials naturally occurring in a fractured condition, or materials naturally occurring of highly angular nature or rough texture.

iii) shall be free from coating of clay, silt or other deleterious material, and

iv) shall meet the requirements listed in Table 502-B.

b) Fine Aggregate

i) shall be all mineral matter passing the sieve designated in the test procedure for each individual test.

ii) shall be clean, tough, durable, moderately sharp, and free from coatings of clay, silt, or other deleterious material, and shall contain no clay balls or other aggregations of fine material.

iii) shall have a maximum mass loss after five cycles of not more than 18% and 23% for pavement Class 1 and 2 respectively when tested in accordance with ASTM Test C-88.

iv) shall have a sand equivalent of not less than 40 when tested in accordance with ASTM Test D-2419

v) shall have a minimum value of 45 when tested according the AASHTO Test T 304, Method "A", - Uncompacted Void Content of Fine Aggregate when determining Fine Aggregate Angularity.

c) Mineral Filler and Mineral Dust

i) Mineral filler shall consist of all matter passing the 0.600-mm sieve and mineral dust shall consist of all mineral matter passing the 0.075-mm sieve.

ii) Mineral filler and mineral dust shall be free from organic matter.

iii) Mineral filler shall be non-plastic when tested in accordance with ASTM Test D-4318.

iv) determination of aggregate properties and gradation for Superpave

d) For the purposes of the mix design for Superpave, the gradation of the design mix and the subsequent Job Mix Formula, shall be of the properties and the gradation limits as specified below and in accordance with the SHRP Standards for a Superpave Level I Mix Design contained in the Asphalt Institute's Superpave Series Publication - Superpave Level I Mix Design SP-2. Changes and/or variations from these limits shall be outlined within the Contract Special Provisions.

i) 90% fractured aggregate with a 12.5mm nominal maximum size, including sufficient manufactured fines to provide fine aggregate angularity. The aggregates must meet all the requirements for angularity, toughness, soundness, deleterious materials, clay content and thin, elongated particles. Design ESALs will be as specified in the

TABLE 502-B – REQUIREMENTS FOR COARSE AGGREGATES

TEST REF. #	PROCEDURES	PAVEMENT CLASSES		
		Super pave	1	2
ASTM C-127	Maximum Water Absorption: % by mass	2	2	2
ASTM C-88	Soundness of Aggregate Maximum loss after 5 cycles	15	15	20
ASTM C-142	Maximum % by mass of clay balls and friable particles	1.0	1.0	1.5
ASTM D-6928	Standard Test Method for Resistance of Coarse Aggregate to Degradation by Abrasion in the Micro-Deval Apparatus	17	17	17
ASTM D-5821	2 Fractured Faces: Minimum % by Mass retained on the 4.75 mm sieve	90	85	70

Contract, or if not specified, 30 million.

ii) A dense graded gradation with a 19.0mm maximum size aggregate restricted to the following:

1) Passing below the following boundaries of the aggregate restricted zone:

PERCENT PASSING BOUNDARIES for 12.5mm Nominal Max. Aggregate Size		
SIEVE SIZE	**Minimum**	**Maximum**
2.36mm	39.1	39.1
1.18mm	25.6	31.6
0.600mm	19.1	23.1
0.300mm	15.5	15.5

2) The design gradation shall be within the following specified control points:

CONTROL POINT (PERCENT PASSING)		
SIEVE SIZE	**Minimum**	**Maximum**
0.075mm	2.0	10.0
2.36mm	28.0	58.0
9.5mm	-	90.0
Nominal Maximum (12.5mm)	90.0	100.0
Maximum (19.0mm)	100.0	-

502.05.03 Supply of Asphalt Cement and Spray Primers - The Contractor shall supply the types and grades of asphalt cement and spray primers as specified in the Contract and in accordance with DBSS 952. The supply of these materials includes, but is not limited to, ordering, scheduling delivery of, receiving, handling, storing, sampling, and testing of the materials and other related work.

The Contractor shall, upon request, supply the Ministry Representative with copies of the asphalt supplier's weigh-bill and records of all asphalt materials received on a daily basis.

502.05.04 Responsibility for Asphalt Mix Design - Preparation and submission of the project asphalt mix designs for review by the Quality Manager is the responsibility of the Contractor. All costs incurred in mix design formulation are the responsibility of the Contractor.

The Contractor shall utilise a qualified registered member of the Association of Professional Engineers and Geoscientists of British Columbia or a qualified, registered member of the Applied Science Technologists and Technicians of British Columbia who shall sign off the asphalt mix design. The Contractor shall also utilise a qualified testing laboratory meeting the requirements of DBSS 502.04.03 and acceptable to the Quality Manager, to assess the aggregate material proposed for use and to carry out the asphalt mix designs.

502.05.05 Requirements for Asphalt Mix Design - The asphalt mix design for Class 1 and Class 2 pavements shall be carried out under the design criteria for Marshall design using the designated equipment and procedures as contained in the Asphalt Institute's Mix Design Methods for Asphalt Concrete MS-2, latest edition.

Asphalt mix designs, Job Mix Formulas and field adjustments made in accordance with DBSS 502.05.09, must all be based on an aggregate gradation meeting the requirements of Table 502-C-1.

The divergence limits, as defined in Table 502-G, when applied to the gradation of the Job Mix Formula, shall not result in a gradation that is outside of the limits as shown in Table 502-C-1.

The Superpave asphalt mix design shall be carried out in accordance with the Superpave Level I Mix Design contained in the Asphalt Institute's Superpave Series Publication - Superpave Level I Mix Design SP-2.

Asphalt mix designs shall be performed using asphalt cement of the penetration grade described in the Contract and Design and which is from the same refinery contracted to supply the asphalt cement for the duration of the project.

Any subsequent change in the asphalt cement supplied by the Contractor will require a new asphalt mix design unless permitted otherwise by the Ministry Representative.

Marshall asphalt mix designs shall be performed to meet the criteria specified in Table 502-C-2 for the class of pavement specified in the Contract and Design.

The asphalt cement content shall be selected such that all aspects of the design criteria are met. After the design asphalt cement content is selected by the Contractor, the Ministry retains the right at the sole discretion of the Ministry, to bump or increase the design asphalt cement content by 0.2 %.

TABLE 502-C-1 – AGGREGATE GRADATION LIMITS

SIEVE SIZE (mm)	PERCENTAGE PASSING BY MASS			
	COARSE ASPHALT MIX	MEDIUM		FINE ASPHALT MIX
		19 mm ASPHALT MIX	16mm ASPHALT MIX	
37.5	100	---		---
25.0	80 – 100	---		---
19.0	60 – 92	100		---
16.0	---	---	100	
12.5	50 – 85	84 – 95	90– 100	100
9.50	40 – 80	73 – 90	73 – 90	90 – 100
4.75	30 – 65	50 – 75	50 – 75	55 – 80
2.36	20 – 50	35 – 57	35 – 57	32 – 64
1.18	15 – 35	26 – 45	26 – 45	24 – 51
0.600	8 – 30	18 – 34	18 – 34	17 – 40
0.300	6 – 22	10 – 26	10 – 26	13 – 29
0.150	3 – 15	6 – 17	6 – 17	8 – 18
0.075	1 – 7	3 – 7	3 – 7	4 – 10

TABLE 502-C-2 – MARSHALL DESIGN CRITERIA

PROPERTY OF LABORATORY COMPACTED PAVING MIXTURE	PAVEMENT CLASS	
	1	2
Number of blows each face of test specimens	75	75
Minimum % Voids in mineral aggregate for maximum particle size: 25.0 mm	--	13
Minimum % Voids in mineral aggregate for maximum particle size: 19.0 mm	14	14
Minimum % Voids in mineral aggregate for 16 mm Asphalt Mix	14.5	14.5
Minimum % Voids in mineral aggregate for maximum particle size: 12.5 mm	15	15
Minimum % Voids in mineral aggregate for maximum particle size: 9.5 mm	16	16
Percentage of air voids in laboratory compacted mixture for: Coarse Mix	--	3 to 6
Percentage of air voids in laboratory compacted mixture for: Medium Mix	2.5 to 4.5	2.5 to 4.5
Percentage of air voids in laboratory compacted mixture for: 16 mm Mix	2.5 to 4.5	2.5 to 4.5
Percentage of air voids in laboratory compacted mixture for: Fine Mix	4 to 6	4 to 6
Minimum Marshall Load, N @ 60°C for 80 – 100 Pen.	9000	7000
Minimum Marshall Load, N @ 60°C for 150 – 200 and 200 – 300 Pen.	7000	6000
Flow index, units of 0.25 mm	8 to 14	8 to 16
Asphalt Film Thickness	Min. 7.5 microns	Min. 7.5 microns
Minimum Index of Retained Stability after immersion in water at 60°C for 24 hours	85%	75%

502.05.06 Asphalt Mix Design Submittals - The Contractor shall submit each asphalt mix design to the Quality Manager for review that it complies with the requirements of the Contract, and shall not proceed until the review is received in writing from the Quality Manager. Also, the Mix Design shall be submitted to the Ministry Representative for information. The Contractor's submissions shall include the following information:

- The gradation of each aggregate to be used in each mixture.
- The percentage by mass of each aggregate to be used in each mixture.
- The design gradation of the combined aggregate for each mixture.
- All asphalt mix design data used in arriving at the final mix designs.
- The design asphalt content expressed as a percentage of the dry weight of the aggregate.

502.05.07 Review of Asphalt Mix Design - The mix design must be reviewed by the Quality Manager prior to commencement of pavement construction.

When the Quality Manager has confirmed in writing that the Contractor's asphalt mix design has been accepted, the accepted asphalt mix design becomes the Job Mix Formula for the start of production.

502.05.08 - Verification of Job Mix Formula - Verification of the Job Mix formula will be carried out by the Contractor during the course of production of the first Lot. The Contractor will provide a Job Mix Formula with supporting verification data, which shall include all the information, but not limited to, the requirements of DBSS 502.05.05 and DBSS 502.05.06.

502.05.09 Field Adjustment of Job Mix Formula - A field adjustment to the Job Mix Formula is defined as a change in the asphalt cement content of the mix, aggregate gradation and/or proportioning of various aggregate sizes, within the specified limits as shown in Table 502 - D without review and acceptance of a new asphalt mix design.

The proposed field adjustment shall be submitted in writing together with supporting documentation to the Quality Manager. The Quality Manager will review the field adjustment for conformance with the contract requirements and notify the Contractor whether or not it is acceptable.

The Contractor's field adjustment to the Job Mix Formula must comply with the Asphalt Mix Design requirements of DBSS 502.05.05 and DBSS 502.05.06. The Contractor shall provide all supporting verification data

After the Job Mix Formula has been established in accordance with DBSS 502.05.09, no field adjustment to that Job Mix Formula will be permitted without prior written acceptance by the Quality Manager. A maximum of two (2) field adjustments may be made to the Job Mix Formula originally derived from the laboratory mix design.

No field adjustment will be acceptable if it results in a change from the Job Mix Formula, for any property, in excess of the maximum adjustment for that property permitted in Table 502-D.

PART B - PAYMENT ADJUSTMENTS

502.11 Density

502.11.01 Coring - The Contractor shall be responsible for providing all core samples for quality testing and payment adjustment purposes. For payment adjustment purposes, the randomly selected locations for the cores shall be supplied by the Ministry Representative to the Contractor, and the cores shall be taken within a 300 mm radius of the specified location. The Contractor shall provide 100mm diameter cores for these purposes. For projects utilizing a Superpave

TABLE 502-D – FIELD ADJUSTMENT OF JOB MIX FORMULA

Job Mix Formula Property	Maximum Field Adjustment
Asphalt cement content, percent	± 0.2
Percentage passing 25.0 mm, 19.0 mm, and 16.00 mm sieves	± 2.0
Percentage passing12.5 mm and 9.5 mm sieves	±2.0
Percentage passing 4.75 mm and 2.36 mm	±1.5
Percentage passing 1.18mm and 0.600mm	± 1.5
Percentage passing 0.300mm and 0.150mm sieves	± 1.5
Percentage passing 0.075 mm sieve	± 0.50

product, 150mm diameter cores shall be required. The Contractor shall prepare the cores prior to the submission by removing all material not representative of the pavement lift to be tested. The Contractor shall deliver these cores to the Ministry Representative on site, within 24 hours of being provided the locations for the coring.

The Contractor shall fill and compact the core holes immediately after the extraction of the cores.

502.11.02 Percent Density - The average in-place density will be determined from core samples of the completed lift of pavement.

One random core sample will be obtained from each Sub-Lot and the test results for the three Sub-Lots will be averaged to determine the percent density for the Lot.

Marshall Mixes

$$\text{Marshall Percent Density} = \frac{\text{In-Place Density of Sample}}{\text{Plant Briquette Density}} \times 100$$

For the determination of Mix Briquette Density, the Contractor shall obtain the samples as outlined in Appendix 2 for each Sub-lot.

From each Sub-lot sample, the Contractor will form three briquettes as per the procedure identified in Table 502-A-3 and tested, and the test results will be averaged to obtain the test value for that sample.

The Mix Briquette Density for a Lot is the average of the test results from the three Sub-lots. The Mix Briquette Density value for the Lot shall be provided to the Ministry Representative prior to the provision of the coring locations to the Contractor.

At the discretion of the Ministry Representative, the Ministry may at any time, perform and use the Ministry results in accordance with procedures as outlined above, to obtain the value for Mix Briquette Density which will replace the values achieved under the Contractor's Quality Management Testing.

Should the initial test results in any Sub-Lot be lower than 92.0%, one additional core will be taken and tested, and that result shall replace and be averaged with the initial result to determine the percent density for the Sub-Lot.

Superpave Mixes

$$\text{Superpave Percent Density} = \frac{\text{In-Place Density of Sample}}{\text{Maximum Theoretical Density (ASTM D-2041)}} \times 100$$

The average in-place density will be determined from 150mm diameter core samples of the completed lift of pavement.

The Maximum Theoretical Density for a Lot is the value calculated defined as the average of the test results from the three sub lots. The Maximum Theoretical Density value for the Lot shall be provided to the Ministry Representative prior to the provision of the coring locations to the Contractor.

502.11.03 Payment Adjustments - The payment adjustment for percent density will be the amount shown in Table 502-E-1 and 502-E-2 for the Sample Mean of the test results for the Lot.

502.11.04 Rejection Limit - The rejection limit for percent density is the limiting value of the Sample Mean as shown in Table 502-E-1 and 502-E-2, beyond which the Sub-Lot is rejected and not paid for.

Acceptance or rejection will be determined for each Sub-Lot. One opportunity will be provided to the Contractor, using non-destructive test methods, to isolate the area of low density and conduct additional compaction providing no damage is incurred to the new pavement. After re-compaction, one additional core will be taken randomly in the area of low density and will replace the core with previous low density and a new value will be calculated. The Ministry Representative shall be advised as to the scheduling of the re-compaction effort prior to the work taking place.

Where the Lot average density is less than 96%, the Contractor shall either overlay or remove and replace the previously placed mix. If the test result for density of a Sub-Lot is outside the acceptance limits, the Sub-Lot is rejected automatically regardless of the values of other acceptance parameters. To minimize the cost of rejection to the Contractor, the Contractor will be provided the opportunity to isolate the area of low density within the Sub-lot. The limits of the low density area must be verified and approved by the Quality Manager, and accepted by Ministry Representative, before remedial work proceeds.

TABLE 502-E-1 – PAYMENT ADJUSTMENTS FOR DENSITY – MARSHALL

% Density Lot Average	Payment Adjustment ($ Per Tonne) for Design Lift Thickness
98.6 Plus	+$1.00
98.1 – 98.5	+$0.75
97.6 – 98.0	+$0.50
97.0 – 97.5	+$0.25
96.6 – 96.9	-$0.50
96.0 – 96.5	-$1.00
Less than 96.0%	Reject

If any Sub-Lot has a density below 96%, the Sub-Lot shall be considered rejectable, and, subject to the Contractor's opportunity to increase the density as described in DBSS 502.11.04, shall be rejected.

TABLE 502-E-2 – PAYMENT ADJUSTMENTS FOR DENSITY - SUPERPAVE

% Density Lot Average	Payment Adjustment ($ Per Tonne) for Design Lift Thickness
95.0 Plus	+$1.00
94.0 – 94.9	+$0.75
93.0 – 93.9	+$0.50
92.0 – 92.9	+$0.25
91.6 – 91.9	-$0.50
91.0 – 91.5	-$1.00
less than 91.0 %	Reject

If any Sub-Lot has a density below 91%, the Sub-Lot shall be considered rejectable, and, subject to the Contractor's opportunity to increase the density as described in DBSS 502.11.04, shall be rejected. Percent Density is based on a comparison to the Maximum Theoretical Density.

Where the Lot average density is less than 91%, the Contractor shall either overlay or remove and replace the previously placed mix.

502.11.05 Payment for Rejected Work Made Acceptable
The payment adjustment for density will be based on testing of the replacement or overlay material where applicable. Where replacement or overlay material does not cover the entire Lot or Sub-Lot, prior tests of the uncovered area will be averaged with new tests on the corrective work.

502.12 Asphalt Content

502.12.01 Determination of Asphalt Content - The asphalt content of the asphalt mix will be determined from loose mix samples obtained from each Sub-Lot (3 per Lot) and tested in accordance with ASTM test procedures listed in Table 502-A-3.

One random sample will be obtained from each Sub-lot as per Appendix 2. The asphalt content of the asphalt mix will be determined using the average of the results obtained from all of the Sub-lot samples. The Actual Asphalt Content of the Lot will be compared to the Job Mix Formula Asphalt Content and the deviation will be used for payment adjustment purposes.

502.12.02 Payment Adjustments - The payment adjustment for asphalt content will be the amount shown in Table 502-F.

502.12.03 Rejection Limit - Rejection limits for asphalt content are the limiting values of the Sample Mean as shown in Table 502-F, beyond which the Lot is rejected and not paid for.

If the asphalt content of a Lot is outside the acceptance limits, the Lot is rejected automatically regardless of the

values of other acceptance parameters.

502.12.04 Payment for Rejected Work Made Acceptable - The payment adjustment for Asphalt Content will be based on testing of the replacement or overlay material where applicable. Where replacement or overlay material does not cover the entire Lot or Sub-Lot, prior tests of the uncovered area will be averaged with new tests on the corrective work.

502.13 Aggregate Gradation

502.13.01 Determination of Aggregate Gradation - Table 502-C-1, Aggregate Gradation Limits specifies the aggregate gradation limits which asphalt mix design, Job Mix Formulas, and field adjustments made in accordance with DBSS 502.05.09, must be based.

Table 502-G shows the limiting values for acceptance and for the determination of payment adjustments. These limits are expressed as departures, on each specified sieve size, from the percentage of material passing that sieve according to the Job Mix Formula. Aggregate gradation will be determined for each Lot from loose mix samples of the completed lift of pavement. One random sample will be obtained from each Sub-lot, and the aggregate gradation for the Lot will be determined by using the average of the 3 sub-lot tests to determine the Sample Mean.

Divergence from the job mix formula gradation shall not vary outside of the limits identified in Table 502-C-1.

502.13.02 Payment Adjustments -

a) Where the Sample Mean for every specified sieve size falls within the applicable limiting value prescribed in Column 3 of Table 502-G, the Lot will be accepted with a payment adjustment of + $0.75 per tonne.

b) Where the Sample Mean for every specified sieve size falls within the applicable limiting value prescribed in Column 2 of Table 502-G, but the Lot is not eligible under (a) above, the Lot will be accepted with no payment adjustment.

c) Where the Sample Mean for every specified sieve size falls within the applicable limiting value prescribed in Column 1 of Table 502-G, but the Lot is not eligible under either (a) or (b) above, the Lot will be accepted with a payment adjustment of -$1.50 per tonne (a deduction of $1.50 per tonne).

502.13.03 Rejection Limit - Rejection limits for aggregate gradation are the limiting values of the Sample Mean for any one, or more, of the specified sieve sizes, as shown in Column 1 of Table 502-G.

If the Sample Mean value for any specified sieve size exceeds the values shown in Table 502-G, or the divergence of the job mix formula gradation is outside the limits of Table 502-C-1, the Lot will be rejected and not paid for, regardless of the values of other acceptance parameters.

Where the Ministry Representative considers it acceptable, a

TABLE 502-F – PAYMENT ADJUSTMENTS FOR ASPHALT CONTENT

Deviation of the Actual Asphalt Content from the Approved Asphalt Content (either above or below) kg/100kg of Dry Aggregate	Payment Adjustment for Asphalt Content $ per tonne	
	Top Lift	Lower Lift
From 0 to 0.15	$0.50	$0.50
From 0.16 to 0.30	$0.00	$0.00
From 0.31 to 0.35	-$1.00	-$1.00
From 0.36 to 0.40	-$2.00	-$2.00
From 0.41 to 0.45	-$3.00	-$3.00
From 0.46 to 0.50	-$4.00	-$4.00
From 0.51 to 0.55	Reject	-$5.00
0.56 and greater	Reject	Reject
For top lift deviations of more than 0.50 the Contractor shall either overlay or remove and replace the Lot.		
For lower lift deviations of more than 0.55, no payment will be made and the Ministry Representative will determine whether removal and replacement is necessary.		

TABLE 502-G – ACCEPTANCE AND PAYMENT ADJUSTMENTS FOR AGGREGATE GRADATION

Sieve Size mm	Divergence from job mix formula grading curve Percentage passing by mass ASTM C-117 and C-136		
	COLUMN 1 Limiting Values for Acceptance Conditional Upon a Payment Adjustment of - $1.50 per tonne of mix (a deduction of $1.50 per t)	COLUMN 2 Limiting Values for Acceptance with No Payment Adjustment	COLUMN 3 Limiting Values for Acceptance with a Payment Adjustment of + $0.75 per tonne of mix
12.5	±7.0	±4.5	±3.5
4.75	±6.0	±4.5	±3.0
0.600	±4.0	±3.5	±2.0
0.075	±1.5	±1.0	±0.75

Any material with a gradation outside the limits set out in Column 1 is reject and the Contractor shall either overlay or, remove and replace the previously placed mix. Reject Bottom Lift shall remain in place, however no payment will be made for this material under the applicable Items

Lot in rejection may be accepted at a minimum deduction of $40 per tonne.

502.14 Material Application Rate

502.14.01 Determination of Material Application Rate - Asphalt mix shall be applied to the roadway at the rate or rates specified in the Design, in conformance with the Contract. Material Application Rates will be determined by the tonnage delivered to the paver as recorded by weigh tickets generated by automated scales, divided by the area covered by the Lot after allowance has been made for entrance letdowns and/or intersections. The Contractor shall provide the material application rate calculations to the Quality Manager at the end of each shift.

502.14.02 Payment Adjustments - The payment adjustment for material application rate will be applied as shown in Table 502-H.

502.14.03 Acceptance and Rejection Limits - The acceptance limit for material application rate is the limiting value of the actual material application rate, expressed as a percentage of the specified material application rate, for the Lot, below which the Lot is rejected, as shown in Table 502-H.

If the application rate of a Lot is outside the acceptance limit, the Lot is rejected automatically regardless of the values of other acceptance parameters.

502.15 Surface Segregation

502.15.01 Determination of Surface Segregation - The finished pavement shall be homogeneous, free from segregation and shall be uniform with respect to surface texture. A segregated area is defined as an area within the driving lanes of the pavement wherein the texture differs visually from the texture of the surrounding pavement.

Due to the nature of variation in asphalt mixes and their compactive characteristics, the definition of degrees of segregation will be established on a project by project basis. The Contractor and the Ministry Representative shall establish through the use of photographs or other mutually agreed tools, the definition of slight, moderate and severe segregation. The Ministry Representative and the Contractor shall evaluate the first two Lots upon completion of the second Lot, after opening to traffic, to confirm the "agreed to" guidelines. Segregation will then be evaluated only when all paving is complete.

The Quality Manager and the Ministry Representative, will observe the finished pavement to evaluate the existence, severity and extent of segregation and other surface defects. The evaluation will be completed prior to the issuance of the Completion Certificate

The rating system for categorizing the severity of top lift surface segregation, and repair methodology is shown in Table 502-I-1.

502.15.02Payment Adjustments - The payment adjustment for top lift segregation will be the applicable amount shown in Table 502-I-2.

502.15.03 Repair - On the final driving pavement surface,

TABLE 502-H – PAYMENT ADJUSTMENTS FOR MATERIAL APPLICATION RATE

Actual Application Rate Expressed as % of Specified Application Rate	Assessment for Application Rate $ per tonne	
	Bottom Lift or Single Lift	**Top Lift of Multiple Lifts**
110 +	-$6.00 for all material in the Lot up to 110% and no payment for product in excess of 110.0%	No payment
106.0 – 109.9	-$4.00 for all material in the Lot	-$6.00 for all material in the Lot up to 106% and no payment for product in excess of 106.0%
105.0 – 105.9		-$4.00 for all material in the Lot
104.0 – 104.9	$0.00	-$2.00 for all material
96.0– 103.9	+$0.50	+ $0.50
94.0 - 95.9	-$1.00	-$1.00
92.0 - 93.9	-$2.00	-$2.00
90.0 - 91.9	-$3.00	-$3.00
85.0 – 89.9	-$5.00	-$5.00
84.9 or lower	Rejected, Mill and Fill, and/or rejected with no remedial work required at the discretion of the Ministry Representative	Rejected, Mill and Fill, Overlay, and/or rejected with no remedial work required at the discretion of the Ministry Representative

all segregation shall be repaired according to Table 502-I-1. The Lot will not be re-evaluated for segregation payment adjustments. The Lot will be re-evaluated for acceptance. Repair shall be to the neat lines and dimensions of the segregated area using a sand cement slurry. The Ministry Representative shall provide the formula for the slurry to the Contractor.

The Contractor shall be responsible for all costs associated with the repair of segregated areas.

Remedial work is required for any occurrences of segregation.

502.16 Smoothness

502.16.01 Determination of Pavement Smoothness - The finished pavement surface shall be tested by the Ministry using a Class I precision rolling profile measuring instrument, to determine the longitudinal profile and compute the International Roughness Index (IRI) in each driving lane. Profiles shall be measured and the IRI calculated in the centre of the lane for each Sub-Lot.

For any Sub-Lot between 50 m and 100 m in length, the IRI value shall be considered representative of a complete Sub-Lot. For any Sub-Lot less than 50 m in length, the IRI value will be combined with the proceeding Sub-Lot IRI value.

The profile shall be measured over the entire length of the pavement exclusive of structures and shoulder areas. Acceleration, deceleration and turning lanes are considered part of the driving lanes and shall be tested in accordance with this provision. For the measuring process, the Contractor shall provide the Ministry Representative a chalk guide line in the centre of the lane immediately prior to measurement.

502.16.02 Auxiliary Lanes - For smoothness testing, sections of the driving lanes that do not fall within the continuous through lanes, such as acceleration lanes, deceleration lanes and turning lanes, and lanes which are less than 1 km in length, shall be treated as follows. The ratio of the section length to the standard Lot length of 1 km shall be determined and the payment adjustment shall be pro-rated on this basis as in the following example:

Length of segment, i.e. 565 metres = 0.565 times the Standard Lot Length of 1000m.

Hence the applicable payment adjustment is 0.565 times the payment adjustment for a 1-km Lot as determined from Table 502-J.

502.16.03 Acceptance Limits – The acceptance limit and payment adjustments applicable to a road for smoothness are as shown in Table 502-J or when so indicated in the Contract, Table 502-J-2. The International Roughness Index (IRI) value, calculated for each Sub-Lot, will be used to determine if the Lot will be accepted, and if so whether it will be subject to any payment adjustment.

502.16.04 Payment Adjustments - Acceptance of any Lot for payment will occur if it contains no obvious defects as per DBSS 502.22 and in the case of top lift pavement only, the pavement smoothness, as reported by the International Roughness Index (IRI) meets the following requirements:

- No Sub Lot shall have a value greater than the "Acceptance Limit" as shown in Table 502-J or when so indicated in the Special Provisions, Table 502-J-2

TABLE 502-I-1 – SEGREGATION – TOP LIFT ONLY – REMEDIATION METHODOLOGY

Segregation Severity	Visual Appearance	Repair Procedures
None	Completely uniform surface texture	N/A
Slight	Matrix of asphalt binder, coarse and fine aggregate exists; visually increased presence of stone sizes	sand cement slurry
Moderate	Significantly more stone than surrounding pavement; matrix of asphalt binder and coated sand particles is reduced	seal coat or sand cement slurry patch or neat hot mix patch or mill and fill patch
Severe	Appears as an area of very stony mix - stone against stone - little or no matrix	remove and replace or overlay to limits defined by Ministry Representative

TABLE 502-I-2 – SEGREGATION – TOP LIFT ONLY – PAYMENT ADJUSTMENTS

Payment Adjustment	Category		
	Slight	Moderate	Severe
+$1,000 per lane km. (If the criteria in all three adjacent columns are met)	0 – 3	0	0
+$500 per lane km. (If the criteria in all three adjacent columns are met)	4-5	0	0
No payment adjustment	6 – 10	0 – 1	0
-$500 per lane km. (If the criteria in any one or more of the adjacent columns applies)	11 – 15	2 – 5	0 – 1
-$1,000 per lane km. (If the criteria in any one or more of the adjacent columns applies)	16+	6+	2+

TABLE 502-J – LOT ASSESSMENT AND PAYMENT ADJUSTMENTS FOR SMOOTHNESS

For the final surface course only, the following payment adjustments shall apply to each Lot.

Lot IRI (m/km)	Payment Adjustment
≤ 0.80	+$2,000
0.81 to 0.90	+$1000
0.91 to 1.00	+$500
1.01 to 1.10	+$200
1.11 to 1.20	0
1.21 to 1.30	-$100
1.31 to 1.40	-$250
1.41 to 1.50	-$600
1.51 to 1.60	-$1,400
1.61 to 1.70	-$2,000
1.71 to 1.80	-$3,000
1.80	**ACCEPTANCE LIMIT.** Reject or corrective work required above this level

The Lot IRI is the average IRI value calculated for the Sub-Lots within the Lot.

TABLE 502-J-2 – LOT ASSESSMENT AND ALTERNATIVE PAYMENT ADJUSTMENTS FOR SMOOTHNESS

For the final surface course only, and only where specifically invoked by the Contract, the following payment adjustments shall apply to each Lot on a road designated by the Ministry for application of alternative smoothness payment adjustments

Lot IRI (m/km)	Payment Adjustment
<1.00	+$2000
1.01 – 1.10	+$1000
1.11 – 1.20	+$500
1.21 – 1.30	+$250
1.31 – 1.40	0
1.41 – 1.60	-$300
1.61 – 1.80	-$750
1.81 – 1.90	-$1500
1.90 – 2.00	-$2500
2.00	**ACCEPTANCE LIMIT.** Reject or corrective work required above this level

The Lot IRI is the average IRI calculated for the Sub-Lots within the Lot

502.16.05 Rejection Limit - When there are one or more Sub-Lots with an IRI greater than the Acceptance Limit shown in table 502-J or when so indicated in the Special Provisions, Table 502-J-2, the Sub-Lot or Sub-Lot is rejected, subject to DBSS 502.16.06. A Sub-Lot for calculation of smoothness is defined in DBSS 502.03.08.

502.16.06 Remedial Work - If the test results on a Sub-Lot of pavement indicate a payment reduction or rejection because of smoothness, the Contractor may propose remedial work to improve the smoothness. Such proposals are subject to the approval of the Quality Manager and the Ministry Representative, but such approval does not imply that the proposed remedy will be successful, and does not reduce the Contractor's responsibility for meeting the acceptance requirements. Grinding may be acceptable, but an overlay may be required. Only one attempt may be made to improve smoothness, and this must be completed within ten (10) calendar days from the time the Contractor receives notification of the original smoothness test results for that Sub-Lot.

Following any attempt to improve the smoothness of a Sub-Lot or Sub-Lots, the Ministry Representative will retest the Sub-Lot or Sub-Lots, and the new results will replace the previous data for the purposes of determining acceptance and payment.

502.16.07 Smoothness Deficiencies - Smoothness deficiencies (bumps and dips) less than 8mm over 3m will not have remedial work required. Individual deficiencies between 8mm and 12mm over 3m will result in a penalty of $200.00 each. Deficiencies exceeding 12mm over 3m will require remedial work.

PART C - ACCEPTANCE AND REJECTION

502.21 Acceptance At Reduced Or Adjusted Payment - Acceptance of any Lot at reduced payment will occur if it contains no obvious defects as per DBSS 502.22, and if:

- the test results for acceptance parameters are such that the Lot or Sub-Lot meets the requirements for acceptance at a reduced payment; and
- the Lot or Sub-Lot is approved in respect of all other requirements; and if
- the Contractor has not notified the Ministry Representative in writing that it will exercise its option to either repair or remove and replace the work, at its own cost, with work meeting the requirements for acceptance at full or increased payment.

502.22 Rejection For Workmanship Defects - The finished surface of any lift shall have a uniform "closed" texture and be free of visible signs of poor workmanship.

Any obvious defects as determined by the Quality Manager or Ministry Representative such as, but not limited to the following, will be cause for automatic rejection of asphalt pavement regardless of the values of any other acceptance parameter:

- individual bumps and dips that exceed 12mm over 3m;
- areas of excess or insufficient asphalt;
- improper matching of longitudinal and transverse joints;
- roller marks;
- tire marks; or
- cracking or tearing.

When asphalt pavement is rejected by reason of obvious defects, the minimum area of rejection will be the actual length of the defect for the full width of the driving lane in which the defect exists.

Rejected work shall be promptly repaired, remedied, overlaid, or removed and replaced in a manner acceptable to the Quality Manager and Ministry Representative. The Contractor shall be responsible for all costs including materials.

No payment will be made for work in any Lot or Sub-Lot, which has been rejected, until the defects have been remedied.

502.23 Appeal Testing

502.23.01 Density, Asphalt Content and Gradation - The Contractor may appeal the results of acceptance testing for density, asphalt content and/or aggregate gradation for any Lot only once. Appeals will only be considered if quality control results support an appeal and can be presented.

Contractor Quality management test results for density which are provided to the Ministry Representative subsequent to the Contractor's receipt of the Ministry's quality audit test results for that Lot will not be considered (when evaluating evidence) for an appeal. The appeal shall be for all tests within the Lot, and there will be no appeal allowed for single tests within a Lot.

Any attempt to improve density on the appealed Lot after the Ministry Representative has tested the Lot for acceptance shall void the appeal and the original test results will apply.

The following procedures will apply for an appeal:

- For Density, Gradation and Asphalt Content, the Contractor shall serve notice of appeal to the Ministry Representative, in writing, within two working days of receipt of the test results;
- The Ministry Representative will arrange for an independent testing laboratory to perform the appeal testing. The personnel employed or testing laboratory retained by the Contractor for quality control testing on the project will not be used for appeal testing;
- The Contractor will sample the pavement and provide the core samples to the independent testing laboratory. The Ministry Representative shall observe the sampling process.
- For density appeals, the original core test results will be discarded and a new sample mean will be obtained from 5 random cores. The appeal agency shall prepare new briquette densities from the previously taken companion samples as per Appendix 2.
- For asphalt content and gradation, the original test results will be discarded. A new sample mean for three new test results will be determined using the companion samples and will be used for acceptance and payment adjustments.

502.23.02 Smoothness - The Contractor may appeal acceptance test results of a Lot only once.

The Ministry Representative will perform, and the Contractor will be given the opportunity to witness, the appeal testing and the new results will be binding on the Contractor and the Ministry.

502.23.03 Segregation - Appeals of segregation ratings will first be handled by a joint review with the Contractor. If consensus cannot be reached then the Ministry Representative shall engage the Joint Committee, comprised of representatives from the Ministry and B.C. Roadbuilders, to assess the area(s) in question.

502.23.04 Application of Appeal Testing Results - For density, gradation and asphalt content appeals, the original test results will be discarded.

For density, five (5) new random cores will be tested within the next three (3) site occupancy days to establish a new Sample Mean and used for acceptance and payment adjustment.

For gradation and asphalt content appeals, companion samples prepared as per Appendix 2 shall be used to establish a new Sample Mean and used for acceptance and payment adjustment.

The new values, thus determined, in all cases, will be binding on the Contractor and the Ministry.

502.23.05 Payment for Appeal Testing - If the new results indicate a change in the payment adjustment in the Contractor's favour, then sampling and testing costs incurred during the appeal procedures for that Lot will be

borne by the Ministry.

If the new results verify that any payment reduction or rejection remains valid for that Lot, then the costs of testing (plus 10% mark-up) incurred during the appeal procedure will be charged to the Contractor.

502.23.06 Time Limits For Appeal Of Test Results – All appeals shall be in writing and submitted within three (3) days of receipt of the applicable test results.

502.24 Overlays As A Corrective Measure - If an overlay is used as a corrective measure on a defective Lot or Sub-Lot, the overlay thickness will be subject to the approval of the Quality Manager and Ministry Representative, but shall not be less than 40mm. In all other respects, the whole overlay will be subject to the same specifications as the pavement being overlaid. Where an overlay is used as a corrective measure in any lane, adjacent lanes shall also be overlaid whether acceptable or not.

Whether the overlay is applied as a corrective measure or is placed over otherwise acceptable pavement in order to match an adjacent lane, acceptability and payment will be determined as follows:

- acceptability, and eligibility for either positive or negative payment adjustment, will be determined entirely on the results of testing and observations conducted on the overlay, regardless of test results that have been obtained on the underlying, overlaid lift of pavement; but
- the quantity to which any payment adjustment is to be applied, will be derived from the tonnage of mix in the underlying, overlaid lift.

PART D - PAYMENT

502.31 General - Applicable payment adjustments (Additions or subtractions as applicable) shall be applied to payments due the Contractor in accordance with Part B of this Section.

The first 1,000 tonnes of asphalt mix production and placement will not be subject to bonus/penalty payment adjustments but will be required to meet minimum Specification requirements for quality and workmanship. The location where this material is to be placed shall be defined in the Contract or as proposed by the Contractor and accepted by the Ministry Representative.

502.31.01 Not Used

502.31.02 Not Used

502.31.03 Not Used

502.31.04 Payment for Acceptable Work - The following end product properties of asphalt pavement will be measured for acceptance:

- Density
- Asphalt Content
- Aggregate Gradation
- Smoothness
- Material Application Rate
- Segregation

502.31.05 Payment for Rejected Work Made Acceptable - When defects have been remedied in Lots or Sub-Lots which had been rejected, payment will be made subject to payment adjustments and penalty assessments and subject to DBSS 502.24.

502.31.06 Payment Adjustment - Payment adjustments resulting from the application of this Section will be effected on each progress payment as follows:

a) For each Lot paid for by the tonne (t), the applicable payment adjustment derived from Tables 502-E (Density), 502-F (Asphalt Content), and 502-G (Gradation), in dollars per tonne, will be expressed as positive in the case of increases and negative in the case of decreases. The algebraic sum of these unit adjustments will then be applied to the payment quantity for the Lot. The resulting amount, in dollars, will be the net payment adjustment, positive or negative, for that Lot.

Payment Adjustments from Table 502-H (Application Rate) shall be computed for each Sub-Lot and then aggregated for the Lot.

The algebraic sum of the net payment adjustments for all such Lots for which payment is authorized on the current progress payment, computed in dollars, shall be the total payment adjustment for density, asphalt content, aggregate gradation, and material application rate for the current progress payment.

b) For each Lot (km) with payment adjustments based on kilometre, the applicable unit adjustment derived from Table 502-I-2 (Segregation) and Table 502-J or 502 J-2 (Smoothness), in dollars per Lot ($/lane km), will be expressed as positive in the case of increases and negative in the case of decreases, and will be the payment adjustment, positive or negative, for that Lot.

The algebraic sum of the payment adjustments for all such Lots for which payment is authorized on the current progress payment, computed in dollars, shall be the total payment adjustment for segregation for the current progress payment.

c) The algebraic sum of the total payment adjustments for density, asphalt content, aggregate gradation, material

application rate, segregation and smoothness, derived in accordance with a) and b) above, shall be the total payment adjustment, positive or negative, in dollars, for all attributes for the current progress payment. This amount shall be added, if positive, or subtracted, if negative, by a single entry in computing the current progress payment.

d) The process set out in a) to c) above will be used in computing each progress payment to which it is applicable.

502.32 Surplus Aggregate In Stockpile – Surplus aggregates in stockpile in Ministry pits is the property of the Ministry and no additional payment will be made.

APPENDIX 1

QUALITY CONTROL REQUIREMENTS AND GUIDELINES

1.01 GENERAL

This Appendix titled "Quality Control Requirements and Guidelines" sets out the requirements and guidelines for the Contractor's Quality Control Plan, in addition to those set out in DBSS 101, DBSS 502.04 and the Contract. Quality Assurance requirements are per the Contract and DBSS 101.

1.02 QUALITY CONTROL PLAN

The Contractor shall prepare and submit a Quality Control Plan for evaluation. The Plan may be operated wholly or in part by a qualified Subcontractor or an independent organization/agency. However, the Quality Control Plan, including compliance with the Plan and its modifications, must remain the responsibility of the Contractor.

The Plan shall address all elements that affect the quality of the hot mix, hot laid asphalt pavement, including but not limited to the following:

- Supply of Asphalt Materials
- Identification of source and proof of quality of aggregates to be supplied;
- Aggregate production and its gradation control;
- Quality of aggregate components;
- Stockpile management;
- Mix Design(s);
- Asphalt plant calibration and proportioning of aggregate components;
- Mixing including asphalt cement content control;
- Process temperature controls;
- Material application rates, placing and finishing;
- Joints;
- Compaction;
- Smoothness;
- Segregation;

The plan shall also include the following:

- The name of the Quality Control testing agency and its proven capability to provide the specific services required for the project.
- The list of dedicated technical staff, if available, (including names, qualifications and relevant experience) and their proposed roles.
- The list of testing equipment available for project work.

The Quality Control Plan shall include the designation of specific personnel to be responsible for specific quality control duties.

- There shall be a Quality Control Manager (QCM) who will be responsible for the development and management of the Quality Control Plan. This person shall be qualified as per the requirements identified in DBSS 502.04.03. This individual shall also be responsible for the signing off of Quality Control Testing and Inspection records. This individual shall also be responsible for approving all quality control staff, implementing and documenting any changes or improvements to the Quality Control Plan.
- There should be a designated Process Control Technician (PCT), with 5 years of related experience, who will ensure that laboratory test results and other quality control practices are used to control the quality of aggregates and other mix components, and to adjust and control mix proportions to meet the mix design(s). The PCT is responsible for ensuring that testing equipment, utilized for proportioning and mixing are calibrated and in good working order. The Plan shall describe how the PCT's duties, including sampling methods and responsibilities are to be accomplished and documented. The Plan should also describe the criteria to be used by the PCT to correct or reject unsatisfactory materials.
- There should also be a Pavement Quality Control Technician (PQT), with 5 years of related experience, who will ensure that delivered materials meet the requirements of the specifications. In addition, the PQT shall be responsible for periodically inspecting all equipment used in placing, finishing, and compacting to assure its proper operating condition and to assure that placing, application rate, finishing, joint construction, and compaction is in conformance with this specification and the contract requirements.

1.03 CONTRACTOR'S RECORD OF QUALITY CONTROL TESTING

Test results should be made on specified forms or charts immediately after completion of each test. These test results are to be made available to the Ministry Representative upon request.

Records of gradation control, both during aggregate production and also during the asphalt mixing operation, should be kept on the form H-295 - Mechanical Analysis of Aggregates.

1.04 MATERIAL APPLICATION RATE

The Contractor shall control the Material Application Rate by monitoring the amount of asphalt mix delivered to the

road against the area covered by checking the application rate minimally every ten loads.

The Contractor shall advise the Ministry Representative in writing on an ongoing basis, and upon request, of the application rate.

1.05 DENSITY

The Contractor should take core samples to determine actual pavement density. At the start of paving, the Contractor should take a minimum of two pavement cores from each Sub-Lot. The Contractor may employ a nuclear densometer to ensure intermediate density control. Two nuclear densities may be determined for each Sub-Lot, based on job mix densities obtained from the most recent plant briquettes.

1.06 OTHER QUALITY CONTROL PROCEDURES

The Contractor may initiate other Quality Control procedures as necessary for ensuring production of a quality product and include them in the Quality Control Plan. Procedures may also be introduced after the start of work as necessary as amendments to the Quality Control Plan.

1.07 QUALITY CONTROL TESTING FREQUENCY

Recommended test frequency requirements for Quality Control are described in the following table:

TABLE 1 – TEST FREQUENCY GUIDELINES

	Test	***Minimum Frequency**
Tests During Aggregate Production	ASTM C-136, Dry Sieve Analysis of Aggregate	- Split Stockpiles: 1 for each stockpile for every 2 hours of production. - One main stockpile: for every 300 tonnes. - Blend Sand: 1 for every 100 tonnes during stockpiling. - Natural filler: 1 for every 50 tonnes during stockpiling.
	ASTM D-5821 Determining the Percentage of Fractured Particles in Coarse Aggregate	Every second coarse aggregate sieve test.
	ASTM C-117 Sieve Analysis of Aggregates by Washing (Field Lab)	1/shift on reduced sample obtained from combined samples from the crusher
Tests During Asphalt Plant Mixing	ASTM 136, Dry Sieve Analysis of Aggregate	1 of combined aggregate (off the belt) every 300 tonnes.
	ASTM C-566 & D-2216, Moisture Content	Aggregate: 2 tests/Lot Asphalt mix: 1 on first Sub-Lot and every second day.
	ASTM C-117 Sieve Analysis of Aggregates by Washing (Field Lab)	1/shift on reduced sample obtained from combined samples from the plant cold feed
	ASTM D-5581, Resistance to Plastic Flow Using Marshall Apparatus.	One set of three briquettes for 1,200 tonnes or Lot, whichever is less.
	Asphalt Extraction Test ASTM D-6307 Ignition Method	One/Lot.
	Penetration of Bituminous Materials ASTM D-5	One per Manufacturer's Batch. Samples should be taken for every 3000 tonnes of mix production
	Viscosity ASTM D-2171	Contractor's Option
Asphalt Products Tests	Tack and Prime	Contractor's option.
Test During Asphalt Paving for Density Testing	ASTM D-2726, D-1188 and D-2041 Test to obtain Briquettes from Asphalt Mix	One 15 kg sample for every Sub-Lot or minimum 1/day for field testing
	Core Samples	At start, two cores for each Sub-Lot. After rolling pattern established, only one core for each Sub-Lot. All cores to be a minimum of 100mm diameter; Superpave mixes shall require minimum 150mm diameter cores

* These are the recommended minimum frequencies and the Contractor is responsible to assess the need to increase testing frequency, where aggregate source is not uniform or any other condition exists that may warrant it.

APPENDIX 2

LOOSE MIX SAMPLE PREPARATION

1. 3 samples per Sub-Lot shall be obtained by the Contractor from the paver augers, at locations and times chosen by the Ministry Representative. The first sample shall be used by the Contractor for Quality Control, the second sample by the Ministry for Quality Audit, with the third retained by the Ministry for potential appeal testing.

2. The samples taken shall meet the requirements of ASTM D-979 Sampling of Bituminous Mixtures, Table 1, Size of Sample

3. The sample size shall be reduced to required laboratory sample size for Marshall and Superpave gyratory briquettes, asphalt content and hot mix gradation determination as outlined below:

I. The sample shall either be reduced using a Riffle splitter or shall be quartered into four approximately equal portions. The two diagonally, opposite quarters shall be combined resulting in two samples. Identify and designate one of the samples as the Quality Companion Sample and set aside. Identify and designate the other resulting sample as the Quality Control Sample. Use the Quality Control Sample for testing.

II. The Quality Control Sample shall be weighed to ensure that the sample so obtained meets the minimum mass required for the ignition test. If the sample does not meet the minimum mass requirements, then the additional materials will be obtained and added to the Quality Control Sample from the Companion Sample. This will be achieved by quartering the Quality Companion sample and adding one quarter of the Quality Companion Sample to the Quality Control Sample.

III. This process is to be repeated for each sample to provide the Quality Audit and Appeal Samples.

4. The third sample for appeal purposes shall be set aside and retained in a Contractor-supplied, suitable container labeled with sample location, date sampled, and project information. These shall be delivered to the Ministry Representative for storing.

5. The areas sampled shall be filled with mix immediately after sampling, and shall receive the same compactive effort as the rest of the mat.

SECTION 503

BASE PREPARATION AND BASE PROCESSING

503.01 Base Preparation - The work shall consist of shaping and compacting the existing granular base course to the designated grade and cross-section.

The existing base course material shall be watered, graded and compacted to provide a non-segregated, tightly-knit surface which conforms to the line, grade and cross sections, as per the Design, to an accuracy of ± 10 mm.

If necessary, the existing surface shall be scarified and bladed to achieve this. Imported materials may be incorporated into the work to aid the Contractor in achieving lines and grades and to backfill failed areas.

a) Failed Areas - Where a failed area is present in the existing base, the Contractor shall excavate and dispose of the material. and backfill using crushed granular base or surfacing aggregate or road-mix (in thicknesses corresponding the excavated depths of like materials) and compacted to the same density as the rest of the base course.

b) Compaction - If watering is required to attain compaction or for any other reason, it shall be applied uniformly from a distributor of the pressure type, equipped with a spray bar mounting nozzles similar to those used on asphalt distributors. The distributor must be provided with a satisfactory means for accurately measuring the quantity of water sprayed. The final surface of the base course shall be compacted to a minimum 100% of the laboratory density obtained by the current ASTM test method D 698.

c) Surplus Aggregate - All surplus loose aggregate arising from the base preparation shall be bladed clear into windrows on the shoulder for later incorporation into the shouldering aggregate or bladed to the side slope and trimmed.

503.02 Base Processing - Where, the base is so deformed, rutted or unsuitable, that it cannot be adequately prepared for priming, the Contractor shall reconstruct the base to the satisfaction of the Designer and Quality Manager.

SECTION 504

PAVEMENT DRAINAGE

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

DESCRIPTION

504.01 Scope - This Section covers the provision of pavement drainage with the construction of necessary asphalt curbing and drainage outlets using cast iron catch basins or asphalt spillways together with pipe or paved outfalls. Concrete curbs, concrete and corrugated steel catch basins, cast iron grates and frames where applicable are specified by DBSS 582.

MATERIALS

504.11 Materials

The Contractor will supply all materials as required in the Contract.

CONSTRUCTION

504.31 Asphalt Curbs

504.31.01 Integral Asphalt Curb - Where specified in the Contract or the Design, asphalt concrete curb integral with the asphalt concrete highway pavement shall be constructed to the dimensions indicated on Drawing SS SP504-01.

504.31.02 Machine-Laid Asphalt Curb - Where specified, machine laid (extruded) asphalt concrete curbs shall be constructed in accordance with Drawing SS SP504-01 and as follows:

i) The type of curb to be placed (Type "A" or Type "B") will be as specified by the Designer.

ii) Prior to construction of the curb, the affected pavement area shall be dry and cleaned of loose or deleterious materials. A tack coat of liquid or emulsified asphalt shall be uniformly applied at a rate of from 0.25 to 0.5 ℓ/m^2, to a width 0.1 m wider than the proposed curb.

iii) The production and transportation of asphalt mix for curb construction shall be in accordance with DBSS 501 and DBSS 951 and shall meet the requirements for Class 1 Fine Mix.

iv) The machine for constructing the curb shall be self-propelled extrusion-type equipment with a material hopper, distributing screw and adjustable curb forming devices capable of placing and compacting the asphalt mix to the required alignment, grade and cross-section in an even homogeneous manner free of surface defects. The asphalt concrete mix shall be handled to provide a smooth, uniform finished curb alignment. Where handforming of curb is required, the finished curb shall conform to all dimensions, alignment, density and finish requirements for machine-laid curb.

504.32 Location of Drainage Outlets - On completion of pavement and curb construction and in conjunction with the installation of any precast concrete barriers, the Contractor shall install or construct drainage outlets at the low point of curves in the vertical alignment and at such other locations designated by the Designer.

Note: The exact low point on the pavement may be determined by allowing water to flow along the surface to the actual lowest point on the pavement adjacent to the curb or barrier. This low point shall be clearly marked on the pavement.

504.33 Excavation - At the locations designated by the Designer, the asphalt pavement and any curbing shall be accurately cut away and the highway shoulder excavated to the dimensions consistent with the installation of Cast Iron Catch Basins (Drawing SS SP504-03) or the construction of Asphalt Spillways (Drawing SS SP504-02). The excavation shall, as nearly as possible, have vertical sides and shall be so done as to cause minimum disturbance of the adjacent compacted base and pavement structures. The outside dimensions of the excavation shall be the smallest compatible with the efficient construction of the drainage facility, as any excess or irregular excavation will be required to be repaired with asphalt mix.

Where applicable, appropriate trench excavation shall be performed to permit the installation and embedment of outlet pipe and associated fittings**504.34 Drainage Outlets**

504.34.01 Cast Iron Catch Basins - Mortar shall be used for bedding catch basins and the mortar shall be composed of 20 MPa concrete.

Bedding shall be a minimum of 125 mm thick to permit accurate setting of the catch basin in mortar at the correct level true to alignment with curbing, barrier or pavement edge as shown on Drawing SS SP504-03. After securely fixing the adaptor, watertight gasket and connecting pipe, the catch basin shall be surrounded with mortar to the full perimeter of the excavation, and the mortar well tamped and worked around the connecting pipe for 300 mm from the side of the catch basin.

The mortar shall be allowed to set prior to backfilling and further work.

504.34.02 Asphalt Spillways - Asphalt spillways shall be constructed to the general dimensions shown on Drawing SS SP504-02 with the necessary handforming in accordance with DB SS 504.31.02 (iv).

504.34.03 Pipe Outfalls General - Except where paved outfalls are specified, corrugated galvanized metal outlet pipes of the required diameter shall be installed, as shown generally on Drawing SS SP504-03. Should the fill material preclude pipe embedment, the pipe shall be secured to the slope by dual angle iron posts placed at 3 m intervals, as shown. The pipe shall be similarly secured at the bend at the base of the slope and at a point approximately 150 mm from its outlet.

The pipe trench shall be backfilled with excavated material and the fill thoroughly compacted.

Riprap of 200 mm minimum dimension shall be placed at the pipe outlet as shown on Drawing SS SP504-03.

504.35 Dishing, Spillways and Completion - The asphalt pavement adjacent to the drainage outlet shall be constructed to the profile and cross-sections indicated for the type of outlet being installed. The asphalt pavement surface shall be so shaped as to channel all collected water to the outlet orifice and no ridges or other obstructions shall be left which may impede the flow of water to the drainage outlet.

The asphalt paving mix shall be compacted to the density as specified by the Designer to provide an integral and impervious drainage area. Where the asphalt pavement abuts a cast iron catch basin box, the contact face of the box shall be painted with liquid or emulsified asphalt prior to placing the asphalt concrete.

On completion, the area shall be cleared and cleaned up to the satisfaction of the Quality Manager and the Ministry Representative.

SECTION 507

PRODUCTION AND STOCKPILING OF ASPHALT MIX

507.01 General - This Section covers the supply of aggregate, mixing it with a specified asphalt material in a central plant and placing the produced mix in a stockpile.

507.02 Aggregate

a) The Contractor shall supply aggregate with gradation wholly within the limits listed in Table 507-A when tested in accordance with ASTM C-136 and ASTM C-117 (wash test).

Once a crushed aggregate gradation has been accepted, the maximum permissible variation of the mean of any four consecutive tests from the accepted gradation curve shall be within the limits specified in Table 507-B.

b) A minimum of 85% by count of all aggregate particles retained on the 4.75 mm sieve shall have at least two fractured faces or shall be naturally angular with sharp edges (ASTM D-5821). If this requirement cannot be met for a specified crushed aggregate type then the Contractor shall, at no additional charge, crush aggregate to the next smaller size or such intermediate size.

c) All aggregates shall be free from coatings of clay, silt or other objectionable matter and shall contain no clay lumps or other aggregations of fine-grained materials.

d) Crushed aggregate shall conform to the following requirements:

- The sand equivalent value shall not be less than 40.
- Micro Deval maximum abrasion loss shall not be greater than 17% as tested under ASTM D6928-03.

e) For testing purposes, fine aggregate shall be all mineral matter passing the 4.75 mm sieve including mineral fillers. It shall consist of natural and/or manufactured material derived by crushing stone, slag or gravel.

f) Mineral filler shall:

- consist of all mineral matter passing the 0.425 mm sieve.
- be dry, free from organic matter, clay particles or lumps.
- be non-plastic when tested in accordance with Test ASTM D 423 and ASTM D 424.

g) Mineral dust shall consist of all mineral matter passing the 0.075 mm sieve.

507.03 Asphalt Material

a) The type of asphalt material to be used will be specified in the Contract.

b) The Contractor will supply the Liquid Asphalts or High Float Emulsified Asphalts, which shall meet the requirements of DB SS 951.

TABLE 507-A GRADATION REQUIREMENTS FOR AGGREGATE

SIEVE SIZE (mm)	PERCENT PASSING (BY MASS)	
	PATCH MIX	PAVING MIX
19.0	---	100
16.0	---	100 - 92
13.2	---	100 - 80
9.5	100	90 - 65
4.75	85 - 55	75 - 45
2.36	65 - 40	60 - 30
1.18	50 - 25	45 - 20
0.300	30 - 15	25 - 10
0.150	18 - 8	17 - 5
0.075	12 - 5	9 - 2

TABLE 507-B MAXIMUM PERMISSIBLE VARIATION OF CRUSHED AGGREGATE GRADATION FROM THE GRADATION CURVE

SIEVE SIZE	VARIATION LIMITS % PASSING
4.75 mm and larger	± 5
1.18 mm to 4.75 mm	± 3.5
0.300 mm to 1.18 mm	± 2
0.075 mm to 0.300 mm	± 1

507.04 Mix Design

a) Unless otherwise specified, the Contractor will carry out the Asphalt Mix design with the aggregate supplied by the Contractor.

b) Not Used

c) Once the design grading curve is established, the maximum permissible variation will be as specified under DB SS 507.02 (a).

507.05 Mixing Requirements

a) Mixing operations shall be performed in a mixing plant capable of producing the asphalt mix within the required specifications.

b) The maximum moisture content of the aggregate immediately prior to mixing with liquid asphalt shall not be over 2% by mass of dry aggregate.

c) The Asphalt storage temperature shall be within the range of 65^0C to 80^0C and a mix temperature within the range of 60^0C to 80^0C.

d) The asphalt content in the mix shall not vary from the design asphalt content by more than 0.3 by mass of total aggregate.

e) The asphalt mix shall not contain more than 1% moisture by mass at the discharge from the plant.

f) The asphalt mix shall be uniform in appearance with all particles thoroughly coated. The mixing time will be increased if the mix is not completely coated with asphalt.

g) The Ministry reserves the right to adjust the asphalt content in the mix up to the maximum of 6% by Mass.

507.06 Hauling - When the haul distance from the mixing site to the stockpile is greater than 1 km, the hauling vehicles shall be equipped with and use a tarpaulin of such size as to completely protect the asphalt mix from the weather and inhibit premature curing of the mix by aeration.

The tarpaulin shall be double quilted canvas or other approved material treated to prevent absorption of the asphalt.

507.07 Stockpiling

a) The exact location of the stockpile site will be determined by the Contractor. The Contractor shall place the cold mix within the boundaries of the designated stockpile site.

b) The Contractor shall prepare the stockpile site, which shall be shaped to a uniform smooth surface and graded to ensure good drainage.

c) Temporary stockpiling of the mix on the ground or in storage bins will be allowed, however, the Contractor will be required to minimize the handling of the mix to prevent premature curing of the mix.

d) If conveyor belts are used for temporary stockpiling or final stockpile construction, the free fall distance from the conveyor belt to the stockpile shall not exceed 2 m.

e) If the stockpile is constructed using a front end loader or end dumping by truck, the mix shall be deposited in layers of approximately 1 m in depth.

f) Plank or protected runways shall be provided for operating equipment on stockpiles to prevent contaminants being tracked onto the stockpile and to minimize consolidation of the mix.

g) The final stockpile shall be constructed in such a manner that height is not more than 4 m.

h) Mixing, hauling or stockpiling will not be permitted during periods of precipitation.

SECTION 508

END PRODUCT SPECIFICATION FOR GRADED AGGREGATE SEAL COAT

PART A – GENERAL

508.01 Preliminary and General - This section describes the materials, equipment, professional standards and the end product requirements for the construction of graded aggregate seal coats. This section shall be read and construed together with all applicable Design Build Standard Specifications. Should this Section conflict with any other Section, this Section shall be take precedence unless otherwise stated in the Special Provisions.

Graded Aggregate Seal shall only be used in the completed Work when explicitly identified in the Contract as an acceptable product.

508.01 General Description of Work - Surface seals shall consist of the application of a selected asphalt emulsion and embedment of the selected aggregate so as to provide a new, uniformly textured surface. This surface shall be impervious to moisture, be skid-resistant and have good riding qualities. There are six classes of Graded Aggregate Seal, as specified in Table 508-A below. These include but are not limited to graded aggregate seals, a coarse or fine sand seal or any other graded aggregate surface treatment outlined in the Contract.

TABLE 508-A GRADATION LIMITS FOR GRADED AGGREGATE SEAL

Sieve size (mm)	% PASSING BY MASS by CLASS					
	A	B	C	Mod C	D	E
19	100	-	-	-	-	-
16	-	100	100	100	-	-
12.5[1]	60-90	60-90	-	-	100	-
9.5	40-80	35-75	30-70	25-55	30-80	100
4.75	20-60	15-50	25-45	7-30	25-45	0-30
0.600	0-25	0-15	5-20	0 10	5-20	0-10
0.075	0-7	0-5	0-3	0-3	0-3	0-3
Ratio[2] (9.5/4.75)	-	-	-	1 - 1.5	-	-

Notes:

1 The Contractor may use either the 12.5 mm or the 13. 2 mm sieve, at the Contractor's option.

2 The "Ratio" is the ratio of the mass of material retained on the 9.5 mm sieve, divided by the mass of material retained on the 4.75 mm sieve.

508.02 Definitions

a) Coarse Aggregate – Coarse aggregate is defined as all material retained on or above the 4.75 mm sieve.

b) Fine Aggregate – Fine aggregate is defined as all materials passing the 4.75 mm.

c) Mineral Filler and Mineral Dust – Mineral filler shall consist of all mineral matter passing the 0.600 mm sieve and the mineral dust shall consist of all mineral matter passing the 0.075 mm sieve.

d) Blinding Sand – Fine aggregate used as large aggregate support.

508.03 Graded Aggregate Seal General Usage

- Class A & B - Double seals on gravel highways
- Class B - Low traffic paved roads
- Class C, Mod C, & D - Major Highways
- Class E - Second seals

PART B - SUPPLY OF MATERIALS

508.11 Asphalt Materials - The Contractor shall supply the asphalt emulsion as specified in the Contract and the Design which shall meet the requirements of DBSS 952.

Asphalt(s) to be used on surface seals may be standard rapid curing cutback asphalt; medium curing cutback asphalts; or anionic or cationic emulsions including high float emulsified asphalts. Types and grades of asphalt binder proposed for use on the Project will be specified in the Contract and the Design.

The Ministry Representative or the Designer may require the use of anti-stripping additives or any other materials formulated to improve the coating and adhesive properties of the asphalt binder to the cover aggregate. Instructions concerning the type of additive to be supplied and the method of incorporation with the asphalt binder will be specified in the Contract or in the Design.

508.12 Aggregates

508.12.01 Mineral Aggregate Supply - Mineral cover aggregate meeting the requirements described hereunder shall be supplied by the Contractor unless specifically described otherwise in the Contract.

The Ministry may supply the aggregates. If the Ministry

does supply aggregates, then the Ministry will supply quality control information, if available, to the Contractor. The Contractor must verify and be satisfied that the Ministry supplied aggregates will meet the quantity and quality requirements prior to submitting a bid.

508.12.02 Mineral Aggregate – Test Methods - Mineral Aggregate will be sampled and tested in accordance with the standard ASTM procedures and tests listed in Table 508-B

TABLE 508-B TEST METHODS FOR MINERAL AGGREGATE

ASTM DESIGNATION	NAME OF TEST
D 75	Sampling stone, slag, gravel, sand, and stone block for use as Highway materials
C 136	Sieve Analysis of Aggregates
C 117	Wash Test of Aggregates
C 142	Clay Lumps and Friable Particles in Aggregates
D 6928	Micro-Deval
D 5821	Fracture Count on Coarse Aggregate
D 4791	Flat and Elongated Particles

508.12.03 Test Requirements - Mineral cover aggregate shall consist of clean, sound, hard, durable particles or fragments of sand, gravel or crushed stones or combination thereof; containing a minimum of thin elongated or flaky pieces. It shall be substantially free from loosely bonded aggregations, clay lumps or other objectionable matter and shall not be markedly hydrophilic in character.

The mineral cover aggregate, when tested by the methods listed in Table 508-B, shall meet the following criteria:

a) Clay Lumps - Coarse aggregate shall not contain more than 0.5% by mass of clay lumps or other aggregation of fine material.

b) Micro-Deval - No individual test shall have a loss factor of more than 20%.

c) Flat and Elongated Particles - The percent flat and or elongated particles must be less than 10% using 4:1 ratio.

d) Fracture - Contrary to ASTM D 5821, only those particles retained on or above the 4.75 mm sieve with two or more fractured faces shall be measured. The minimum % by mass, for coarse aggregate shall be as follows:

- Classes A & B = 75%
- Classes C, Mod C, & D = 85%
- Class E = not applicable

508.12.04 Graded Aggregate Seal Gradation.- .Gradations shall meet the requirements stated in Table 508-A for each Class of graded aggregate seal.

a) Variation Limits – For all Classes of seal except Class E, once the grading curve is established within the above limits, the maximum permissible variation of the mean of any four consecutive tests from the grading curve shall be within the tolerances specified in Table 508-C

TABLE 508-C VARIATION LIMITS

Sieve Size (mm)	Maximum Permissible Tolerance % Passing by Mass
4.75 to 19.0	± 5.0
0.600	± 3.5
0.075	± 1.0

These tolerances do not waive the requirement that the running average of four consecutive tests must be maintained at all times inside the limits specified in Table 508-A. Should the Contractor not be able to maintain the grading within the variation limits specified above they shall be required to construct a second stockpile. This does not alleviate the Contractor from consistently maintaining the grading curve within the tolerances indicated in the gradation curves.

508.12.05 Aggregate Characteristics

a) Coarse Aggregate Coarse aggregate shall consist of crushed stone or gravel, or a combination thereof; or materials naturally occurring in a fractured condition; or materials naturally occurring of highly angular nature or rough texture.

b) Fine Aggregate Fine aggregate shall be clean, hard, durable, moderately sharp, and free from coatings of clay, silt or other deleterious material and shall contain no clay balls or other aggregations of fine material. All aggregate passing the 4.75 mm sieve shall be non-plastic.

c) Mineral Filler and Mineral Dust Mineral filler and mineral dust shall be free from organic matter.